Fiscal Year:	FY 2011	**Task Last Updated:**	FY 06/27/2011
PI Name:	Tafrov, Stefan Ph.D.		
Project Title:	DNA Damage Clusters in Human Cell Transformation Induced by Single or Multiple Space Radiation Ion Exposures		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Upton	**State:**	NY
Zip Code:	11973-5000	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
Start Date:	12/16/2009	**End Date:**	05/30/2011
No. of Post Docs:	3	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date now 5/30/2011 because PI changed to Paul Wilson, per C. Guidry/JSC (Ed., 6/27/2011) NOTE: Received no cost extension to 2/29/2012, per C. Guidry/JSC (Ed., 6/21/2011)		
Key Personnel Changes/Previous PI:	ED. NOTE (6/27/2011): PI Paul Wilson took over the project from Stefan Tafrov on 5/31/2011, thus changing the end date on this (Tafrov's) grant to 5/30/2011. Tafrov had previously replaced, in December 2009, former PI Betsy Sutherland, who passed away in October 2009. SEE SUTHERLAND for reports from FY2007-FY2009; SEE WILSON for reports for FY2011 and beyond.		
COI Name (Institution):			
Grant/Contract No.:	NNJ07HC73I		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The space radiation environment poses long and short-term risks to human travelers. During a 3-year Mars mission, the nucleus of each cell in an astronaut's body will be hit by ~400 protons and 0.4 high energy, high atomic number (HZE) ions, plus high levels of protons during Solar Particle Events (SPEs). Predicting and mitigating cancer risk from complex radiation fields in outer space is essential for manned exploration of Mars, as well as long-term habitation of the moon. This requires understanding the roles and importance of damages that induce critical changes, cellular repair responses and potentially-oncogenic cellular alterations. Thus, understanding the biological consequences of multiple particle exposures is essential for assessing radiation risks of space travel. Our research focuses on understanding the consequences to human cells of exposure to the levels and kinds of radiations present on the moon and in deep space. We use a human primary cell neoplastic transformation model, in which irradiated human cells acquire growth properties of human cancer cells. We found that exposure to a moderate proton dose followed by a HZE particle gave higher transformation levels than the sum of the two exposures, indicating non-linear effects of multiple radiation exposures. We are continuing our studies of exposing human cells to multiple space ions including protons at low doses, in SPE-simulating proton fields and in combination with HZE ions. Cell survival and transformation levels are measured. Space radiation induces complex DNA damages affecting both DNA strands/double strand breaks (DSBs) and clusters of damaged bases. We are measuring the induction, repair and levels of unrepaired clusters and DSBs, in order to correlate these data to cell killing and transformation. We developed methods for quantifying DNA bistranded clusters in human cells irradiated with low doses of low LET radiation. We showed that Fe ions, X- and X-rays induce clusters in cells with a linear dose-response and that clusters are repaired via repair-intermediate DSBs and abasic clusters. We recently developed a method for measuring DNA cluster levels in cells exposed to relevant occupational levels of low LET radiation, (mSv doses). However, the induction and repair of clusters in cells exposed to simulated complex radiation fields, and the effects of such radiation on induction of potentially oncogenic damage are unknown. We are initiating studies to determine whether levels of slowly-repaired or unrepaired, persistent clusters correlate with the human cell neoplastic transformation frequencies observed with mixed field radiations. We are determining whether conditions or agents that reduce the levels or types of endogenous or radiation-induced clusters also reduce transformation levels, thus suggesting approaches to mitigation of cancer risk during space travel. Melatonin, a human hormone, effectively scavenges radiation-induced hydroxyl radicals that damage DNA. Our preliminary data indicate that melatonin effectively reduces cell transformation by HZE ions. We propose to test its effectiveness in reducing biological damage induced in human cells by single or multiple space particles.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research is directed to revealing the types of DNA damage inflicted by different types of radiation. Also we are studying the intermediate steps in the anchorage-independent growth of cells exposed to radiation. Both approaches could help to develop strategies for better protection of the healthy cells and higher successful rate for the treatment of the tumor cells with radiation.
Task Progress:	ED. NOTE (6/27/2011): PI Paul Wilson took over the project from Stefan Tafrov on 5/31/2011, thus changing the end date on this (Tafrov's) grant to 5/30/2011. Tafrov had previously replaced, in December 2009, former PI Betsy Sutherland, who passed away in October 2009. SEE SUTHERLAND for reports from FY2007-FY2009; SEE WILSON for reports for FY2011 and beyond. Information per C. Guidry/JSC.
Bibliography Type:	Description: (Last Updated: 03/11/2021)